

Ferric Arenosols:

Ferric Arenosols are sandy soils that contain a ferric horizon – a subsurface layer enriched with iron oxides, often reddish or yellowish in color.

The iron accumulation gives the soil its distinct hue and may cause hardening (ferricrete) when strongly developed.

Drainage:

- Arenosols are **very well drained**, because sand has large pores. Water infiltrates quickly.
- **Con:** water retention (especially available water for plants) is very low.

Depth:

- Arenosols are usually deep, often deeper than 100 cm, a calcic horizon might appear somewhere below 15 cm and extend for some thickness (often 15 cm or more).
- The depth of “usable soil” depends on how deep the sand profile is before hitting an impermeable layer, cemented calcic horizon, or bedrock.

Workability: Because of the sandy texture, Arenosols are **easy to work** mechanically: they are light, not very cohesive, don’t require much force to till etc. This is good for tillage.

- **Con:** they’re prone to wind erosion, shifting, collapse of structure under rain, etc. Also seedling establishment can be difficult in very loose sandy soils.
- If the ferric horizon hardens into lateritic ironstone, cultivation becomes very difficult and root penetration is limited.

Nutrient Richness & Cation Exchange Capacity (CEC):

- Very low fertility and CEC due to dominance of sand and low organic matter.
- Nutrients like nitrogen, phosphorus, and potassium are often deficient.
- Iron is abundant, but excess iron oxides can bind phosphorus, making it unavailable to plants.

pH:

- Generally acidic to slightly neutral, depending on leaching and degree of weathering.
- Tends to lack the buffering effect found in calcic soils

Management:

- Increase organic matter inputs (mulches, compost, cover crops) to improve fertility and water retention.
- Use fertilizers in split applications to reduce leaching, especially for N and K.
- Apply phosphorus in forms less easily fixed by iron (e.g., rock phosphate or banded P).
- Maintain ground cover to reduce erosion.
- Choose crops with a deeper or more aggressive rooting system.